Indian Crops Guide 2025-26

Practical package of practices for key cereals, millets, pulses, oilseeds and commercial crops

Audience: Indian farmers, FPOs, extension workers and agri-entrepreneurs
Coverage: Rice, wheat, maize, barley, jowar, bajra, ragi, small millets, tur, gram, urad, moong, lentil, major oilseeds, sugarcane, cotton, jute/mesta, tobacco, sannhemp and guar

Prepared: 12 January 2026 (reference year: 2025-26)

This document provides indicative agronomic guidance. Input doses and chemical use must be adapted to local recommendations, soil test results, irrigation availability, and label directions. Always follow safety precautions and legal requirements.

How to use this guide

This guide covers key food grains, pulses, oilseeds and commercial crops commonly tracked in India’s agricultural statistics. It is written as a practical field reference: seasons, major states, crop duration, seed rate, irrigation, fertilizer, key pests/diseases, and harvest practices. Recommendations are indicative ranges; always refine using **soil testing** and your local Krishi Vigyan Kendra (KVK) / State Agricultural University (SAU) advisories.

Indian cropping seasons (broadly):
Kharif: sowing with monsoon (Jun-Jul), harvest Sep-Nov. **Rabi:** sowing post-monsoon (Oct-Nov), harvest Feb-Apr. **Summer/Zaid:** short crops in Feb-Jun under irrigation.

Note on 2025-26 “1st Advance Estimates”: Government production estimates are released in multiple rounds (1st advance, 2nd advance, etc.) and get revised as more field and market information becomes available. Use them as a directional snapshot, not final numbers.

Unit conversions mentioned in many reports:
Cotton production is often reported in **bales**, where **1 bale = 170 kg**.
Jute, sannhemp and mesta fibre production may be reported in bales, where **1 bale = 180 kg**.

Practical planning checklist (before you sow)

- **Soil test:** pH, organic carbon, available N-P-K and key micronutrients (Zn, B, S).
- **Water plan:** rainfed vs irrigated; check pump/drip readiness; plan critical-stage irrigations.
- **Seed:** use certified seed; ensure germination; treat seed to prevent seed-borne diseases.
- **Nutrition:** plan basal + split fertilizer; add organic manures where possible; consider sulphur for oilseeds.
- **Weeds:** decide early strategy (pre-emergence, interculture, mulching) to protect first 30-45 days.
- **IPM:** weekly scouting; pheromone traps where relevant; spray only when thresholds are crossed.
- **Market:** check local MSP/procurement, buyers, and storage; plan harvest timing and drying.

Season and crop selection (quick guide)

If irrigation is limited or rainfall is uncertain, prefer **millets** and **pulses**. If you have assured irrigation and market linkages, **rice**, **wheat**, **sugarcane** and **cotton** may be viable. Always match crop choice to soil type (drainage), labour availability, and price risk.

Water need (relative)	Typical crops	Notes
High	Rice (puddled), Sugarcane	Adopt water-saving methods (AWD/DSR, drip, mulching) to reduce cost and risk.
Medium	Wheat, Maize, Cotton, Groundnut, Sunflower, Mustard	Focus on critical-stage irrigation and split fertilizers.
Low	Jowar, Bajra, Ragi, Small millets, Pulses, Guar	Best for drylands; improves soil health; often lower input risk.

Crop groups and coverage

Some entries in agricultural tables are **groups** (e.g., “Cereals”, “Nutri/Coarse Cereals”, “Shree Anna/Nutri Cereals”, “Total Pulses”, “Total Food Grains”, “Total Oilseeds”). These are **aggregations** of the individual crops. This guide provides practical packages for the main individual crops listed.

Shree Anna / Nutri Cereals: a policy term commonly used for millets and other nutri-cereals (e.g., jowar, bajra, ragi and small millets).

Cereals and millets (food grains)

Crop	Seasons	Major states (indicative)	Main use
Rice (Paddy)	Kharif (main), Rabi (boro/late), Summer (where irrigation available)	West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Telangana, Odisha…	Staple food grain
Wheat	Rabi (main)	Uttar Pradesh, Madhya Pradesh, Punjab, Haryana, Rajasthan, Bihar…	Staple food grain (atta/maida)
Maize	Kharif, Rabi (in irrigated/southern/western), Summer (spring maize in north)	Karnataka, Madhya Pradesh, Maharashtra, Bihar, Uttar Pradesh, Rajasthan…	Food, feed, poultry, starch industry, ethanol
Barley	Rabi	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh, Punjab, Bihar…	Feed and malt (brewing), health foods
Jowar (Sorghum)	Kharif, Rabi (post-rainy), Summer (fodder/irrigated)	Maharashtra, Karnataka, Telangana, Madhya Pradesh, Andhra Pradesh, Rajasthan…	Staple in dry regions
Bajra (Pearl Millet)	Kharif, Summer (in some irrigated tracts)	Rajasthan, Uttar Pradesh, Haryana, Gujarat, Maharashtra, Karnataka…	Highly drought tolerant
Ragi (Finger Millet)	Kharif (main)	Karnataka, Tamil Nadu, Uttarakhand, Odisha, Andhra Pradesh, Telangana…	Highly nutritious (calcium/iron)
Small Millets (Kodo, Kutki/Little, Foxtail, Barnyard, Proso etc.)	Kharif (main)	Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh, Telangana, Karnataka…	Climate-resilient

Pulses

Crop	Seasons	Major states (indicative)	Main use
Tur / Arhar (Pigeonpea)	Kharif (main; long/medium duration)	Maharashtra, Madhya Pradesh, Uttar Pradesh, Karnataka, Gujarat, Telangana…	Protein-rich dal
Gram / Chickpea	Rabi (main)	Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka, Telangana…	Major protein pulse
Urad (Black Gram)	Kharif, Rabi (in south), Summer (after rabi harvest with irrigation)	Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra, Andhra Pradesh, Telangana…	Short-duration pulse
Moong (Green Gram)	Kharif, Rabi (south), Summer (highly popular after wheat/rice)	Rajasthan, Madhya Pradesh, Maharashtra, Uttar Pradesh, Andhra Pradesh, Telangana…	Very short duration
Lentil (Masoor)	Rabi	Madhya Pradesh, Uttar Pradesh, Bihar, West Bengal, Jharkhand, Rajasthan…	Protein pulse

Oilseeds

Crop	Seasons	Major states (indicative)	Main use
Groundnut (Peanut)	Kharif, Rabi (south), Summer (irrigated)	Gujarat, Rajasthan, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra…	Edible oil, confectionery, fodder (haulms)
Castor	Kharif (main)	Gujarat, Rajasthan, Telangana, Andhra Pradesh	Industrial oil (lubricants, chemicals, pharmaceuticals)
Sesamum (Til)	Kharif, Rabi (south), Summer (irrigated)	Gujarat, Rajasthan, Madhya Pradesh, Uttar Pradesh, Bihar, Odisha…	Edible oil
Nigerseed (Ramtil)	Kharif (main; also rabi in some hills)	Madhya Pradesh, Chhattisgarh, Odisha, Jharkhand, Maharashtra, Karnataka…	Edible oil in tribal/hilly areas
Soybean	Kharif (main)	Madhya Pradesh, Maharashtra, Rajasthan, Chhattisgarh, Telangana, Karnataka	Edible oil + protein meal for poultry/feed
Sunflower	Kharif, Rabi, Summer	Karnataka, Andhra Pradesh, Telangana, Maharashtra, Tamil Nadu, Bihar…	High-quality edible oil
Rapeseed & Mustard	Rabi (main)	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh, Bihar, West Bengal…	Major edible oil crop
Linseed (Flaxseed/Alsi)	Rabi	Madhya Pradesh, Uttar Pradesh, Chhattisgarh, Bihar, Odisha, Maharashtra	Industrial and edible oil
Safflower	Rabi	Maharashtra, Karnataka, Telangana, Andhra Pradesh, Madhya Pradesh	Edible oil

Commercial and fibre crops

Crop	Seasons	Major states (indicative)	Main use
Sugarcane	Kharif planting (autumn/spring plantings depending on region; harvested year-round)	Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Bihar, Gujarat…	Sugar, jaggery, ethanol
Cotton	Kharif (main)	Gujarat, Maharashtra, Telangana, Andhra Pradesh, Punjab, Haryana…	Fibre for textiles
Jute	Kharif (main)	West Bengal, Bihar, Assam, Odisha, Meghalaya, Tripura	Natural fibre (sacks, geotextiles)
Mesta (Kenaf & Roselle)	Kharif	West Bengal, Assam, Odisha, Bihar, Andhra Pradesh	Fibre similar to jute
Tobacco	Kharif (varies by type; flue-cured often rabi in some regions)	Gujarat, Andhra Pradesh, Karnataka, Telangana, Uttar Pradesh, Bihar…	Industrial/cash crop
Sannhemp (Crotalaria juncea)	Kharif	Uttar Pradesh, Madhya Pradesh, Maharashtra, Odisha, Andhra Pradesh, Telangana…	Fibre + green manure
Guarseed (Clusterbean / Guar)	Kharif	Rajasthan, Haryana, Gujarat, Punjab	Guar gum for food and industry

Crop-wise practical packages

Each crop factsheet below is structured the same way for easy comparison: seasons, states, duration, seed rate, water, fertilizer (indicative), weed and pest risks, and harvest guidance. Doses are ranges; use soil test and local recommendations to finalize.

Rice (Paddy)

Crop group	Cereals
Seasons in India	Kharif (main), Rabi (boro/late), Summer (where irrigation available)
Major producing states	West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Telangana, Odisha, Bihar, Chhattisgarh, Tamil Nadu, Assam
Typical duration	90-160 days (variety dependent)
Water requirement	High. Typical 1000-2000 mm total depending on method; save water via AWD, SRI, DSR, laser leveling.
Expected yield (indicative)	2.5-6.5 t/ha grain (higher under irrigated/high-input); straw 4-8 t/ha.
Main uses	Staple food grain; straw for fodder; by-products (bran, husk) used for feed, oil and energy.

Agro-climate and soil

Warm and humid; 20-35°C. Suitable from heavy clay to loam with good water holding; pH ~5.0-8.0.

Seed and sowing

- **Sowing window:** Kharif: nursery May-Jun; transplant Jun-Jul. Rabi (boro): Nov-Dec in eastern India. Summer: Jan-Feb in irrigated tracts.
- **Seed rate:** Transplanted: 25-35 kg/ha (nursery); Direct seeded: 40-60 kg/ha (line) or 60-80 kg/ha (broadcast).
- **Spacing / plant population:** Transplanting: 20×15 cm or 20×10 cm (1-2 seedlings/hill). DSR line: 20 cm rows.

Nutrient management (fertilizer)

Typical range (soil-test based): N 80-150 kg/ha, P2O5 40-60, K2O 40-60. Add Zn in deficient soils; incorporate FYM/compost.
Always prioritize soil testing and split doses to reduce losses.

Water management

High. Typical 1000-2000 mm total depending on method; save water via AWD, SRI, DSR, laser leveling.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Pre-emergence herbicides (as per label) in DSR; hand/cono weeding in transplanted. Keep first 40 DAS weed-free.
- **Pests/diseases:** Stem borer, leaf folder, planthoppers; blast, bacterial leaf blight, sheath blight. Use resistant varieties + IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when 80-85% panicles straw-colored; grain moisture ~20-22% then dry to 13-14%.

- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Prefer soil testing; avoid excess nitrogen (lodging, pests). In saline/alkaline soils use tolerant varieties and gypsum where advised.

Wheat

Crop group	Cereals
Seasons in India	Rabi (main)
Major producing states	Uttar Pradesh, Madhya Pradesh, Punjab, Haryana, Rajasthan, Bihar, Gujarat, Maharashtra
Typical duration	110-150 days
Water requirement	Medium. 4-6 irrigations typical; critical stages: crown root initiation (CRI ~20-25 DAS), booting, flowering, milking.
Expected yield (indicative)	3-6 t/ha (irrigated); 2-4 t/ha (rainfed).
Main uses	Staple food grain (atta/maida); straw for fodder; strong market linkage (procurement in many states).

Agro-climate and soil

Cool season crop; optimum 10-25°C. Well-drained loam to clay loam; pH 6.0-8.5.

Seed and sowing

- **Sowing window:** Oct-Nov (timely); late sowing up to mid-Dec depending on region; avoid very late to reduce heat stress.
- **Seed rate:** 100-125 kg/ha (timely sown); 125-150 kg/ha (late sown).
- **Spacing / plant population:** Row spacing 20-22.5 cm; seed depth 4-5 cm.

Nutrient management (fertilizer)

Typical: N 100-150 kg/ha, P2O5 40-60, K2O 20-40; split N (basal + CRI + later). Add S in deficient soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium. 4-6 irrigations typical; critical stages: crown root initiation (CRI ~20-25 DAS), booting, flowering, milking.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weed control (25-35 DAS). Use integrated approach: stale seedbed, mechanical weeding, herbicides as per label.
- **Pests/diseases:** Aphids, termites; rusts (yellow/brown), smut, blight. Use resistant varieties; timely sowing; seed treatment.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when grains hard and moisture ~20%; thresh and dry to 12-13%.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.

- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Heat at grain filling reduces yield; choose suitable variety and sow on time. Residue management helps soil health.

Maize

Crop group	Cereals
Seasons in India	Kharif, Rabi (in irrigated/southern/western), Summer (spring maize in north)
Major producing states	Karnataka, Madhya Pradesh, Maharashtra, Bihar, Uttar Pradesh, Rajasthan, Andhra Pradesh, Telangana, Tamil Nadu, Gujarat
Typical duration	90-120 days (grain); 60-90 days (fodder)
Water requirement	Medium. Critical stages: knee-high, tasseling-silking, grain filling. Avoid moisture stress at silking.
Expected yield (indicative)	3-8 t/ha grain; 8-15 t/ha green fodder.
Main uses	Food, feed, poultry, starch industry, ethanol; stover for fodder.

Agro-climate and soil

Warm; optimum 18-32°C; sensitive to frost and waterlogging. Well-drained loam; pH 5.5-7.5.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Rabi: Oct-Nov (south/west); Summer: Feb-Mar (north) with irrigation.
- **Seed rate:** 15-25 kg/ha (hybrids vary); maintain target plant population.
- **Spacing / plant population:** 60×20 cm (approx) or 45×20 cm depending on hybrid; 1 plant/hill.

Nutrient management (fertilizer)

High N demand. Typical: N 120-200, P2O5 50-60, K2O 40-60; split N (basal + knee-high + tasseling). Add Zn.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium. Critical stages: knee-high, tasseling-silking, grain filling. Avoid moisture stress at silking.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30-45 DAS; use pre-emergence + interculture/earthing up.
- **Pests/diseases:** Fall armyworm, stem borer; downy mildew, leaf blights. Use seed treatment, pheromone traps, timely sprays if threshold crossed.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when husk turns brown and grains hard (black layer in some hybrids).
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Use balanced fertilization and proper plant population. Drain excess water in heavy rains.

Barley

Crop group	Cereals
Seasons in India	Rabi
Major producing states	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh, Punjab, Bihar, Himachal Pradesh
Typical duration	110-140 days
Water requirement	Low to medium. 2-3 irrigations; critical at tillering and heading.
Expected yield (indicative)	2.5-5 t/ha
Main uses	Feed and malt (brewing), health foods; hardy in saline/alkaline and drought-prone areas.

Agro-climate and soil

Cool and dry; more tolerant than wheat to salinity and drought. Well-drained loam to sandy loam; pH 6-9.

Seed and sowing

- **Sowing window:** Oct-Nov
- **Seed rate:** 75-100 kg/ha
- **Spacing / plant population:** 22.5 cm rows; depth 4-5 cm

Nutrient management (fertilizer)

Typical: N 60-80, P_2O_5 30-40, K_2O 20; avoid excess N (lodging, poor malt quality).
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium. 2-3 irrigations; critical at tillering and heading.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weed control like wheat.
- **Pests/diseases:** Aphids; leaf rust, powdery mildew. Resistant varieties + timely management.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest at full maturity when spikes yellow; dry to safe moisture.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Suitable for marginal lands; good option under limited irrigation.

Jowar (Sorghum)

Crop group	Millets (Coarse cereals)
Seasons in India	Kharif, Rabi (post-rainy), Summer (fodder/irrigated)
Major producing states	Maharashtra, Karnataka, Telangana, Madhya Pradesh, Andhra Pradesh, Rajasthan, Tamil Nadu, Gujarat
Typical duration	90-120 days (grain); 45-60 days (fodder)
Water requirement	Low to medium; critical at booting/flowering and grain filling if irrigated.
Expected yield (indicative)	1.5-4 t/ha grain; 8-20 t/ha fodder.
Main uses	Staple in dry regions; excellent fodder; drought tolerant; suited for climate resilience.

Agro-climate and soil

Warm; tolerant to drought and high temperature. Well-drained loam to clay loam; pH 6-8.5.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Rabi: Sep-Oct (post-rainy); Summer: Jan-Feb (fodder).
- **Seed rate:** 8-12 kg/ha (grain); 25-40 kg/ha (fodder broadcast/line).
- **Spacing / plant population:** 45×15 cm (grain) or 30×10 cm (high density) depending on variety.

Nutrient management (fertilizer)

Typical: N 60-90, P_2O_5 30-40, K_2O 20-30; split N.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; critical at booting/flowering and grain filling if irrigated.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30 DAS; inter-cultivation.
- **Pests/diseases:** Shoot fly, stem borer; grain mold, anthracnose. Seed treatment, timely sowing, resistant varieties.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when grains hard; avoid bird damage near maturity.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Post-rainy sorghum quality depends on conserved soil moisture.

Bajra (Pearl Millet)

Crop group	Millets (Coarse cereals)
Seasons in India	Kharif, Summer (in some irrigated tracts)
Major producing states	Rajasthan, Uttar Pradesh, Haryana, Gujarat, Maharashtra, Karnataka, Tamil Nadu, Madhya Pradesh
Typical duration	75-100 days
Water requirement	Low; 1-2 protective irrigations can boost yield. Avoid waterlogging.
Expected yield (indicative)	1.5-3.5 t/ha grain; good fodder yield.
Main uses	Highly drought tolerant; nutritious; fodder and grain.

Agro-climate and soil

Hot and dry; performs on sandy soils; pH 5.5-8.5.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Summer: Feb-Mar (with irrigation).
- **Seed rate:** 3-5 kg/ha (hybrids); 5-6 kg/ha (open pollinated).
- **Spacing / plant population:** 45×15 cm or 60×15 cm depending on hybrid.

Nutrient management (fertilizer)

Typical: N 60-80, P2O5 30-40; K generally low need. Apply FYM in sandy soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; 1-2 protective irrigations can boost yield. Avoid waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding and interculture.
- **Pests/diseases:** Downy mildew, blast; shoot fly. Use resistant hybrids, seed treatment.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when earheads dry; thresh and dry grain.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Suitable for arid/semi-arid regions; improves resilience.

Ragi (Finger Millet)

Crop group	Millets (Nutri-cereals / Shree Anna)
Seasons in India	Kharif (main)
Major producing states	Karnataka, Tamil Nadu, Uttarakhand, Odisha, Andhra Pradesh, Telangana, Maharashtra, Chhattisgarh
Typical duration	90-120 days
Water requirement	Low to medium; mostly rainfed; critical irrigation at flowering if available.
Expected yield (indicative)	1.2-3.5 t/ha grain.
Main uses	Highly nutritious (calcium/iron); good for health foods; straw is valued fodder.

Agro-climate and soil

Warm with moderate rainfall; grows in poor soils. Well-drained loam to lateritic; pH 5-7.5.

Seed and sowing

- **Sowing window:** Jun-Jul (kharif); in south can be Aug-Sep depending on rains.
- **Seed rate:** Line sowing: 8-10 kg/ha; transplanting: 4-5 kg/ha for nursery.
- **Spacing / plant population:** Line: 30×10 cm; transplanting: 25×10 cm.

Nutrient management (fertilizer)

Typical: N 40-60, P_2O_5 20-30, K_2O 20-30; apply FYM/compost; lime in acidic soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; mostly rainfed; critical irrigation at flowering if available.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30-40 days; interculture/hand weeding.
- **Pests/diseases:** Blast, leaf spot; stem borer. Use clean seed, resistant varieties, proper spacing.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest earheads in 2-3 pickings or cut whole plant at maturity; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Good crop for smallholders; strong demand in nutrition markets.

Small Millets (Kodo, Kutki/Little, Foxtail, Barnyard, Proso etc.)

Crop group	Millets (Nutri-cereals / Shree Anna)
Seasons in India	Kharif (main)
Major producing states	Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Jharkhand, Uttarakhand
Typical duration	70-110 days
Water requirement	Low; mainly rainfed.
Expected yield (indicative)	0.8-2.5 t/ha grain.
Main uses	Climate-resilient; high fibre/minerals; suited for drylands and tribal regions.

Agro-climate and soil

Warm; tolerant to drought; grows on marginal lands.

Seed and sowing

- **Sowing window:** Jun-Jul; can extend to Aug in some areas.
- **Seed rate:** 6-10 kg/ha (species dependent)
- **Spacing / plant population:** 25-30 cm rows; thin to 8-10 cm within row.

Nutrient management (fertilizer)

Typical: N 30-50, P_2O_5 15-25, K_2O 15-25; add FYM; micronutrients if deficient.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; mainly rainfed.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding is critical; hand weeding/interculture.
- **Pests/diseases:** Generally hardy; blast/leaf spots can occur; manage with resistant varieties and timely sowing.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Cut at maturity; dry, thresh, and store grain dry.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Often benefits from intercropping with pulses; processing/cleaning improves market value.

Tur / Arhar (Pigeonpea)

Crop group	Pulses
Seasons in India	Kharif (main; long/medium duration)
Major producing states	Maharashtra, Madhya Pradesh, Uttar Pradesh, Karnataka, Gujarat, Telangana, Andhra Pradesh, Bihar
Typical duration	140-200 days (variety dependent)
Water requirement	Low to medium; mostly rainfed; one irrigation at flowering/pod fill improves yield if drought.
Expected yield (indicative)	1-2.5 t/ha (rainfed to irrigated).
Main uses	Protein-rich dal; improves soil fertility through nitrogen fixation; good intercrop with soybean/cotton/maize.

Agro-climate and soil

Warm; 20-30°C; sensitive to waterlogging. Well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Jun-Jul with onset of monsoon; avoid late sowing in wilt-prone areas.
- **Seed rate:** 12-20 kg/ha (depending on spacing and variety).
- **Spacing / plant population:** 75×20 cm (long duration) or 60×20 cm (medium); adjust for intercrop.

Nutrient management (fertilizer)

As legume: low N (starter 15-20 kg/ha) + P2O5 40-60 + K2O 20; apply Rhizobium + PSB inoculation.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; mostly rainfed; one irrigation at flowering/pod fill improves yield if drought.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** First 45 DAS weed-free; interculture/earthing up.
- **Pests/diseases:** Pod borer (Helicoverpa), pod fly; wilt, sterility mosaic. IPM (pheromone traps, biocontrol, need-based sprays).
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when 75-80% pods mature; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Rotate crops to reduce wilt; avoid continuous pigeonpea.

Gram / Chickpea

Crop group	Pulses
Seasons in India	Rabi (main)
Major producing states	Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka, Telangana, Andhra Pradesh, Gujarat
Typical duration	90-120 days (desi); 100-140 days (kabuli)
Water requirement	Low. Mostly rainfed; 1-2 irrigations at branching and pod formation if moisture stress.
Expected yield (indicative)	1-3 t/ha (rainfed to irrigated).
Main uses	Major protein pulse; good for dry rabi; fixes nitrogen; valuable straw/haulm for fodder.

Agro-climate and soil

Cool and dry; 15-25°C. Well-drained loam to clay loam; avoids waterlogging; pH 6-8.

Seed and sowing

- **Sowing window:** Oct-Nov (timely) after rains; late sowing reduces yield.
- **Seed rate:** Desi: 60-80 kg/ha; Kabuli: 80-120 kg/ha (large seed).
- **Spacing / plant population:** 30×10 cm (desi) or 45×10 cm (kabuli); depth 8-10 cm in residual moisture.

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 40-60 + K2O 20; add S 20 kg/ha in deficient soils; Rhizobium inoculation.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low. Mostly rainfed; 1-2 irrigations at branching and pod formation if moisture stress.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** One hoeing at 25-30 DAS; pre-emergence herbicide options as per label.
- **Pests/diseases:** Pod borer, cutworm; wilt, dry root rot, Ascochyta blight. Use resistant varieties + seed treatment + IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when plants dry and pods brown; avoid shattering; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Avoid heavy irrigation; prefers conserved soil moisture.

Urad (Black Gram)

Crop group	Pulses
Seasons in India	Kharif, Rabi (in south), Summer (after rabi harvest with irrigation)
Major producing states	Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra, Andhra Pradesh, Telangana, Tamil Nadu, Odisha, Bihar
Typical duration	70-90 days
Water requirement	Low; 2-3 irrigations for summer crop; avoid waterlogging.
Expected yield (indicative)	0.8-1.5 t/ha
Main uses	Short-duration pulse; fits in rotations; fixes nitrogen; good for catch crop.

Agro-climate and soil

Warm; 25-35°C; well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Rabi (south): Oct-Nov; Summer: Mar-Apr (irrigated).
- **Seed rate:** 15-20 kg/ha (line) ; 20-25 kg/ha (broadcast).
- **Spacing / plant population:** 30×10 cm or 45×10 cm depending on variety.

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 30-40 + K2O 20; Rhizobium + PSB; micronutrients if needed.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; 2-3 irrigations for summer crop; avoid waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30 DAS; one hand weeding/interculture.
- **Pests/diseases:** Yellow mosaic virus, leaf crinkle; thrips/whitefly; powdery mildew. Use tolerant varieties and vector control.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when 70-80% pods mature; pick in 2-3 rounds to reduce shattering.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Summer urad is profitable with irrigation and timely harvesting.

Moong (Green Gram)

Crop group	Pulses
Seasons in India	Kharif, Rabi (south), Summer (highly popular after wheat/rice)
Major producing states	Rajasthan, Madhya Pradesh, Maharashtra, Uttar Pradesh, Andhra Pradesh, Telangana, Karnataka, Odisha, Bihar, Punjab (summer)
Typical duration	60-75 days
Water requirement	Low; summer moong needs 3-4 light irrigations; avoid standing water.
Expected yield (indicative)	0.7-1.6 t/ha
Main uses	Very short duration; improves soil; good in spring/summer windows; high market demand.

Agro-climate and soil

Warm; prefers dry weather at maturity; well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Rabi (south): Oct-Nov; Summer: Mar-Apr (north) with irrigation.
- **Seed rate:** 12-20 kg/ha
- **Spacing / plant population:** 30×10 cm or 45×10 cm.

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 30-40; K 20 where needed; Rhizobium inoculation. Foliar DAP/urea spray sometimes used at flowering.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; summer moong needs 3-4 light irrigations; avoid standing water.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** First 25-30 days critical; one hoeing/hand weeding.
- **Pests/diseases:** YMV, powdery mildew; whitefly/thrips. Use resistant varieties and IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when pods black/brown; do 2 pickings if uneven maturity.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Summer moong adds income and breaks cereal-cereal cycle.

Lentil (Masoor)

Crop group	Pulses
Seasons in India	Rabi
Major producing states	Madhya Pradesh, Uttar Pradesh, Bihar, West Bengal, Jharkhand, Rajasthan, Chhattisgarh
Typical duration	110-130 days
Water requirement	Low; one irrigation at flowering/pod fill if needed.
Expected yield (indicative)	1-2 t/ha
Main uses	Protein pulse; fits in rainfed rabi; tolerant to low fertility; improves soil.

Agro-climate and soil

Cool; 15-25°C. Light to medium loam; pH 6-8; sensitive to waterlogging.

Seed and sowing

- **Sowing window:** Oct-Nov
- **Seed rate:** 30-40 kg/ha (small seed) to 40-50 kg/ha (large).
- **Spacing / plant population:** 25×10 cm or 30×10 cm

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 40-60; Rhizobium; S in deficient areas.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; one irrigation at flowering/pod fill if needed.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** One hand weeding at 25-30 DAS; herbicides where used.
- **Pests/diseases:** Aphids; rust, wilt. Use clean seed and resistant varieties.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when pods dry; thresh and dry grain.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Often grown as paira/utera (relay) in rice fallows in eastern India.

Groundnut (Peanut)

Crop group	Oilseeds
Seasons in India	Kharif, Rabi (south), Summer (irrigated)
Major producing states	Gujarat, Rajasthan, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra, Telangana, Madhya Pradesh
Typical duration	100-140 days
Water requirement	Medium; critical at flowering/pegging and pod filling; avoid water stress and waterlogging.
Expected yield (indicative)	1.5-3.5 t/ha pods (higher in irrigated).
Main uses	Edible oil, confectionery, fodder (haulms); fixes nitrogen (legume).

Agro-climate and soil

Warm; 20-30°C; light sandy loam ideal for pegging; pH 6-7.5; avoid waterlogging.

Seed and sowing

- **Sowing window:** Kharif: Jun-Jul; Rabi: Oct-Nov (south); Summer: Jan-Feb (irrigated).
- **Seed rate:** Kernel 100-125 kg/ha (bunch) and 80-100 kg/ha (spreading) depending on seed size; use bold, viable seed.
- **Spacing / plant population:** 30×10 cm (bunch) or 45×15 cm (spreading).

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 40-60 + K2O 20-40; Gypsum 200-500 kg/ha at flowering for calcium & sulphur; Rhizobium inoculation.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; critical at flowering/pegging and pod filling; avoid water stress and waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 45 days; pre-emergence + interculture.
- **Pests/diseases:** Leaf miner, white grub; tikka (leaf spots), rust, bud necrosis. Seed treatment + timely sprays/biocontrol.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when 70-75% pods mature; pull plants, dry pods to 8-10% moisture.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Gypsum and proper drying are key for yield and quality.

Castor

Crop group	Oilseeds (Industrial)
Seasons in India	Kharif (main)
Major producing states	Gujarat, Rajasthan, Telangana, Andhra Pradesh
Typical duration	150-210 days
Water requirement	Low to medium; sensitive to waterlogging; 2-3 irrigations can help in dry spells.
Expected yield (indicative)	1-2.5 t/ha seed.
Main uses	Industrial oil (lubricants, chemicals, pharmaceuticals).

Agro-climate and soil

Warm and dry; tolerant to drought; well-drained soils; pH 6-8.5.

Seed and sowing

- **Sowing window:** Jun-Jul
- **Seed rate:** 8-10 kg/ha
- **Spacing / plant population:** 90×60 cm or wider (variety dependent).

Nutrient management (fertilizer)

Typical: N 60-80, P_2O_5 40, K_2O 20-40; split N. Add FYM.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; sensitive to waterlogging; 2-3 irrigations can help in dry spells.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed control early; interculture.
- **Pests/diseases:** Semilooper, capsule borer; wilt. Use tolerant hybrids and IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest racemes as capsules mature; multiple pickings.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Handle seed safely; castor cake used as organic manure (non-edible).

Sesamum (Til)

Crop group	Oilseeds
Seasons in India	Kharif, Rabi (south), Summer (irrigated)
Major producing states	Gujarat, Rajasthan, Madhya Pradesh, Uttar Pradesh, Bihar, Odisha, West Bengal, Andhra Pradesh, Telangana, Tamil Nadu
Typical duration	80-100 days
Water requirement	Low; 1-2 irrigations in summer; avoid waterlogging and heavy rains at flowering.
Expected yield (indicative)	0.6-1.2 t/ha seed.
Main uses	Edible oil; high-value seeds for sweets and export; drought tolerant.

Agro-climate and soil

Warm; prefers well-drained light soils; pH 5.5-8.

Seed and sowing

- **Sowing window:** Kharif: Jul; Rabi: Oct-Nov (south); Summer: Feb-Mar (irrigated).
- **Seed rate:** 3-5 kg/ha (line) ; 5-6 kg/ha (broadcast).
- **Spacing / plant population:** 30×10 cm or 45×10 cm; thin to proper stand.

Nutrient management (fertilizer)

Typical: N 30-50, P_2O_5 20-30, K_2O 20; S 20 kg/ha in deficient soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; 1-2 irrigations in summer; avoid waterlogging and heavy rains at flowering.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** First 20-30 days critical; one thinning + weeding.
- **Pests/diseases:** Leaf/web caterpillar; phyllody; Alternaria leaf spot. Use clean seed and vector control.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when leaves yellow and capsules begin to turn; cut and stack to avoid shattering; thresh carefully.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Timely harvest is crucial due to shattering.

Nigerseed (Ramtil)

Crop group	Oilseeds
Seasons in India	Kharif (main; also rabi in some hills)
Major producing states	Madhya Pradesh, Chhattisgarh, Odisha, Jharkhand, Maharashtra, Karnataka, Uttarakhand
Typical duration	90-120 days
Water requirement	Low; rainfed.
Expected yield (indicative)	0.4-0.8 t/ha
Main uses	Edible oil in tribal/hilly areas; bird feed; hardy on marginal lands.

Agro-climate and soil

Cooler kharif climates; tolerates poor soils; well-drained soils.

Seed and sowing

- **Sowing window:** Jun-Jul
- **Seed rate:** 4-6 kg/ha
- **Spacing / plant population:** 30×10 cm

Nutrient management (fertilizer)

Typical: N 20-40, P2O5 20; FYM helps.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; rainfed.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding required.
- **Pests/diseases:** Generally hardy; leaf spots may occur.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when heads turn brown; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Good option for low-input farmers in hilly/tribal tracts.

Soybean

Crop group	Oilseeds (also pulse-like legume)
Seasons in India	Kharif (main)
Major producing states	Madhya Pradesh, Maharashtra, Rajasthan, Chhattisgarh, Telangana, Karnataka
Typical duration	90-110 days
Water requirement	Medium; mostly rainfed; ensure drainage. Critical at flowering and pod fill.
Expected yield (indicative)	1.5-3 t/ha
Main uses	Edible oil + protein meal for poultry/feed; improves soil nitrogen; strong processing industry.

Agro-climate and soil

Warm with well-distributed rainfall; sensitive to waterlogging. Well-drained loam; pH 6-7.5.

Seed and sowing

- **Sowing window:** Jun-Jul (with monsoon onset).
- **Seed rate:** 60-80 kg/ha (variety/seed size dependent).
- **Spacing / plant population:** 45×5-7 cm or 30×10 cm depending on variety; target 3-4 lakh plants/ha.

Nutrient management (fertilizer)

Starter N 20 + P2O5 40-60 + K2O 20; Rhizobium inoculation essential; S 20 kg/ha often beneficial.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; mostly rainfed; ensure drainage. Critical at flowering and pod fill.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30-40 days; pre-emergence + mechanical weeding.
- **Pests/diseases:** Girdle beetle, semilooper; rust, yellow mosaic, charcoal rot. IPM and resistant varieties.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when leaves drop and pods dry; avoid delay (shattering).
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: In waterlogged fields adopt raised beds and proper drainage.

Sunflower

Crop group	Oilseeds
Seasons in India	Kharif, Rabi, Summer
Major producing states	Karnataka, Andhra Pradesh, Telangana, Maharashtra, Tamil Nadu, Bihar, Haryana
Typical duration	85-110 days
Water requirement	Medium; critical at bud formation, flowering, seed fill.
Expected yield (indicative)	1-2.5 t/ha
Main uses	High-quality edible oil; fits multiple seasons; good for pollinators.

Agro-climate and soil

Warm; sensitive to frost and waterlogging. Well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Kharif: Jul; Rabi: Oct-Nov; Summer: Jan-Feb.
- **Seed rate:** 5-8 kg/ha
- **Spacing / plant population:** 60×30 cm or 45×30 cm depending on hybrid.

Nutrient management (fertilizer)

Typical: N 60-90, P_2O_5 40-60, K_2O 40; boron may be needed in deficient soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; critical at bud formation, flowering, seed fill.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weed control; inter-cultivation.
- **Pests/diseases:** Head borer, sucking pests; downy mildew, Alternaria. Manage with resistant hybrids and IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when back of head turns yellow and bracts brown; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Bee activity improves seed set; avoid insecticide sprays during peak flowering hours.

Rapeseed & Mustard

Crop group	Oilseeds
Seasons in India	Rabi (main)
Major producing states	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh, Bihar, West Bengal, Assam, Punjab
Typical duration	110-150 days
Water requirement	Low to medium; 2-3 irrigations; critical at flowering and pod formation.
Expected yield (indicative)	1-2.5 t/ha
Main uses	Major edible oil crop; mustard cake for feed/manure; good in rabi rotations.

Agro-climate and soil

Cool; optimum 10-25°C. Well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Oct-Nov
- **Seed rate:** 4-6 kg/ha (line); 6-8 kg/ha (broadcast).
- **Spacing / plant population:** 45×15 cm or 30×10 cm depending on variety.

Nutrient management (fertilizer)

Typical: N 80-120, P_2O_5 40-60, K_2O 20-40; S 20-40 kg/ha is important; boron in deficient soils.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; 2-3 irrigations; critical at flowering and pod formation.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30-40 days; herbicide options as per label.
- **Pests/diseases:** Mustard aphid, painted bug; Alternaria blight, white rust. Use tolerant varieties, timely sowing, need-based sprays.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when 75% pods turn yellow; avoid shattering; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Sulphur is key for oil content; use gypsum/SSP as sources.

Linseed (Flaxseed/Alsi)

Crop group	Oilseeds
Seasons in India	Rabi
Major producing states	Madhya Pradesh, Uttar Pradesh, Chhattisgarh, Bihar, Odisha, Maharashtra
Typical duration	110-130 days
Water requirement	Low; 1-2 irrigations if needed.
Expected yield (indicative)	0.8-1.5 t/ha
Main uses	Industrial and edible oil; omega-3 rich seed; fibre in some types.

Agro-climate and soil

Cool season; well-drained loam; pH 6-8.

Seed and sowing

- **Sowing window:** Oct-Nov
- **Seed rate:** 20-25 kg/ha
- **Spacing / plant population:** 25 cm rows

Nutrient management (fertilizer)

Typical: N 40-60, P2O5 20-40; S beneficial.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; 1-2 irrigations if needed.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding.
- **Pests/diseases:** Aphids; wilt, rust. Manage with seed treatment and resistant varieties.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when capsules turn brown; dry and thresh carefully.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Avoid lodging by balanced nitrogen and proper spacing.

Safflower

Crop group	Oilseeds
Seasons in India	Rabi
Major producing states	Maharashtra, Karnataka, Telangana, Andhra Pradesh, Madhya Pradesh
Typical duration	120-150 days
Water requirement	Low; mostly rainfed; one irrigation at flowering can help.
Expected yield (indicative)	0.8-1.5 t/ha
Main uses	Edible oil; hardy in dry rabi; petals used as natural dye/tea.

Agro-climate and soil

Cool, dry; deep-rooted and drought tolerant. Well-drained soils; pH 6-8.5.

Seed and sowing

- **Sowing window:** Oct-Nov (post-rainy moisture) in peninsular India.
- **Seed rate:** 8-10 kg/ha
- **Spacing / plant population:** 45×20 cm

Nutrient management (fertilizer)

Typical: N 40-60, P_2O_5 20-30; FYM beneficial.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; mostly rainfed; one irrigation at flowering can help.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding and interculture.
- **Pests/diseases:** Aphids; wilt, rust. Manage with IPM and tolerant varieties.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when heads dry; thorns require careful handling.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Works well in dryland rotations; deep ploughing helps.

Sugarcane

Crop group	Commercial crops
Seasons in India	Kharif planting (autumn/spring plantings depending on region; harvested year-round)
Major producing states	Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Bihar, Gujarat, Andhra Pradesh, Telangana, Punjab, Haryana
Typical duration	10-18 months (region and type)
Water requirement	High. Frequent irrigations; adopt drip + fertigation to save 30-50% water; avoid waterlogging.
Expected yield (indicative)	60-120 t/ha cane (wide range by region/irrigation).
Main uses	Sugar, jaggery, ethanol; by-products (bagasse, pressmud) for energy and manure; steady cash crop where water available.

Agro-climate and soil

Warm; 20-35°C; sensitive to frost. Deep, fertile, well-drained loam/clay loam; pH 6-8.5.

Seed and sowing

- **Sowing window:** North: spring (Feb-Mar) and autumn (Sep-Oct). South/west: year-round with best windows. (Follow local).
- **Seed rate:** Sett requirement varies: about 6-8 t/ha of cane setts (2-3 bud setts).
- **Spacing / plant population:** Row spacing 75-120 cm; paired row/trench systems common.

Nutrient management (fertilizer)

High nutrient demand; typical: N 150-250, P2O5 60-80, K2O 60-120; split doses; add FYM/pressmud; micronutrients as needed.
Always prioritize soil testing and split doses to reduce losses.

Water management

High. Frequent irrigations; adopt drip + fertigation to save 30-50% water; avoid waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Critical in first 3-4 months; mulching, interculture, herbicides as per label.
- **Pests/diseases:** Early shoot borer, top borer; red rot, smut. Use healthy seed cane, resistant varieties, clean cultivation.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest at maturity (10-12 months north; 12-18 months south) based on brix; avoid delay after maturity.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Ratoon management important: stubble shaving, gap filling, balanced nutrients.

Cotton

Crop group	Fibre crops
Seasons in India	Kharif (main)
Major producing states	Gujarat, Maharashtra, Telangana, Andhra Pradesh, Punjab, Haryana, Rajasthan, Madhya Pradesh, Karnataka, Tamil Nadu
Typical duration	150-220 days (Bt hybrids often longer)
Water requirement	Medium to high; critical at square formation, flowering, boll development; drip improves water-use efficiency.
Expected yield (indicative)	Lint yield varies widely; seed cotton 1.5-4.5 t/ha; (Production often reported in bales: 1 bale=170 kg lint equivalent as per note).
Main uses	Fibre for textiles; cottonseed oil and cake; major cash crop.

Agro-climate and soil

Warm; 21-30°C; sensitive to waterlogging and frost. Deep black soils to alluvial; pH 6-8.5.

Seed and sowing

- **Sowing window:** May-Jun (irrigated pre-monsoon) or Jun-Jul (rainfed) depending on region.
- **Seed rate:** Depends on hybrid/spacing: ~1.5-2.5 kg/ha (delinted seed) for hybrids; check label.
- **Spacing / plant population:** 90×45 cm or 120×45 cm (Bt hybrids); closer for compact varieties.

Nutrient management (fertilizer)

Typical: N 100-180, P_2O_5 40-60, K_2O 60-90; split N & K; add Zn/B in deficient soils; apply FYM.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium to high; critical at square formation, flowering, boll development; drip improves water-use efficiency.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 60 days; use pre-emergence + interculture + mulching.
- **Pests/diseases:** Sucking pests (jassid, whitefly), bollworms; pink bollworm; wilt. Follow IPM, refuge and resistance management for Bt.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Multiple pickings of open bolls; avoid moisture; proper drying and storage.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Timely pest scouting is essential; avoid indiscriminate insecticide use to prevent resistance.

Jute

Crop group	Fibre crops
Seasons in India	Kharif (main)
Major producing states	West Bengal, Bihar, Assam, Odisha, Meghalaya, Tripura
Typical duration	100-150 days
Water requirement	Medium; moisture needed during early growth; avoid waterlogging.
Expected yield (indicative)	2-3.5 t/ha fibre.
Main uses	Natural fibre (sacks, geotextiles); biodegradable; good cash crop in humid eastern India.

Agro-climate and soil

Warm and humid; 24-35°C; needs 150-200 cm rainfall. Alluvial, well-drained but moisture-retentive soils; pH 6-7.5.

Seed and sowing

- **Sowing window:** Mar-May (pre-monsoon) depending on region.
- **Seed rate:** Capsularis: 5-7 kg/ha; Olitorius: 3-5 kg/ha (line sowing).
- **Spacing / plant population:** Row 25-30 cm; thin plants to 5-7 cm.

Nutrient management (fertilizer)

Typical: N 60-80, P2O5 30-40, K2O 40; apply in splits; FYM beneficial.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; moisture needed during early growth; avoid waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weeding and thinning are critical.
- **Pests/diseases:** Hairy caterpillar, stem weevil; soft rot. Manage with clean cultivation and need-based control.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest at 120-130 days (early flowering) for best fibre quality; retting in clean water 10-20 days.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Production often reported in bales (1 bale=180 kg for jute as per note). Clean retting improves fibre grade.

Mesta (Kenaf & Roselle)

Crop group	Fibre crops
Seasons in India	Kharif
Major producing states	West Bengal, Assam, Odisha, Bihar, Andhra Pradesh
Typical duration	120-150 days
Water requirement	Medium; tolerates temporary waterlogging better than jute.
Expected yield (indicative)	1.5-3 t/ha fibre.
Main uses	Fibre similar to jute; tolerant to waterlogging; used for sacks, ropes, paper pulp.

Agro-climate and soil

Warm and humid; grows in a wider range including waterlogged areas.

Seed and sowing

- **Sowing window:** Apr-Jun
- **Seed rate:** 8-12 kg/ha
- **Spacing / plant population:** 30 cm rows; thin plants.

Nutrient management (fertilizer)

Typical: N 60-80, P_2O_5 30-40, K_2O 40; FYM helpful.
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; tolerates temporary waterlogging better than jute.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weeding and thinning early.
- **Pests/diseases:** Stem weevil, leaf spots; manage with IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest at early flowering; retting similar to jute.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Reported in bales (1 bale=180 kg) per note.

Tobacco

Crop group	Commercial crops
Seasons in India	Kharif (varies by type; flue-cured often rabi in some regions)
Major producing states	Gujarat, Andhra Pradesh, Karnataka, Telangana, Uttar Pradesh, Bihar, Odisha
Typical duration	100-150 days (field) + curing time
Water requirement	Medium; controlled irrigation; avoid water stress and waterlogging.
Expected yield (indicative)	1.5-2.5 t/ha cured leaf (wide variation).
Main uses	Industrial/cash crop; requires strict curing/quality management; regulated marketing.

Agro-climate and soil

Warm; well-drained sandy loam to loam; pH 5.5-7.5; low chloride preferred for quality.

Seed and sowing

- **Sowing window:** Depends on type and region; commonly nursery then transplant at onset of suitable season (follow local board/extension).
- **Seed rate:** Very small seed; raised nursery; transplanting; seed ~0.5-1 kg sufficient for large area via nursery.
- **Spacing / plant population:** Typically 90×60 cm (varies).

Nutrient management (fertilizer)

Crop-specific; often requires N 80-120 with careful timing; K is important for quality; avoid excess N late (quality loss).
Always prioritize soil testing and split doses to reduce losses.

Water management

Medium; controlled irrigation; avoid water stress and waterlogging.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Early weed control; interculture.
- **Pests/diseases:** Aphids, budworm; black shank, mosaic. Nursery hygiene + IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Leaf picking in rounds based on maturity; curing (air/flue/sun) crucial.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Follow local regulations and buyer specifications; quality depends more on curing than yield.

Sannhemp (Crotalaria juncea)

Crop group	Fibre crops / Green manure
Seasons in India	Kharif
Major producing states	Uttar Pradesh, Madhya Pradesh, Maharashtra, Odisha, Andhra Pradesh, Telangana, Gujarat
Typical duration	60-120 days (green manure 45-60 days)
Water requirement	Low to medium; mostly rainfed.
Expected yield (indicative)	Fibre 1-2 t/ha; green biomass 15-25 t/ha.
Main uses	Fibre + green manure; improves soil organic matter; suppresses nematodes; used for ropes and paper.

Agro-climate and soil

Warm; grows in many soils; pH 6-8.

Seed and sowing

- **Sowing window:** Jun-Jul
- **Seed rate:** 25-35 kg/ha (fibre); 40-50 kg/ha (green manure broadcast).
- **Spacing / plant population:** 30 cm rows (fibre) or broadcast for green manure.

Nutrient management (fertilizer)

Low; starter P helpful; benefits from Rhizobium.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low to medium; mostly rainfed.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Fast growing; minimal weeding after establishment.
- **Pests/diseases:** Generally hardy.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- For fibre: harvest at early flowering; retting for fibre extraction. For green manure: incorporate before flowering.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Production reported in bales (1 bale=180 kg) per note for sannhemp fibre.

Guarseed (Clusterbean / Guar)

Crop group	Commercial crops (Industrial legume)
Seasons in India	Kharif
Major producing states	Rajasthan, Haryana, Gujarat, Punjab
Typical duration	90-120 days
Water requirement	Low; rainfed; 1-2 irrigations in dry spells increase yield.
Expected yield (indicative)	0.8-1.8 t/ha seed.
Main uses	Guar gum for food and industry; drought hardy; also fodder; improves soil nitrogen.

Agro-climate and soil

Arid/semi-arid; warm; thrives on sandy loam; pH 7-8.5.

Seed and sowing

- **Sowing window:** Jun-Jul
- **Seed rate:** 15-20 kg/ha
- **Spacing / plant population:** 45×15 cm

Nutrient management (fertilizer)

Starter N 15-20 + P2O5 20-30; Rhizobium inoculation.
Always prioritize soil testing and split doses to reduce losses.

Water management

Low; rainfed; 1-2 irrigations in dry spells increase yield.
Use mulching, laser leveling/raised beds, and micro-irrigation where feasible.

Weed, pest and disease management

- **Weeds:** Weed-free first 30-40 days; interculture.
- **Pests/diseases:** Aphids, jassids; powdery mildew. IPM.
- **IPM basics:** clean seed, timely sowing, crop rotation, field scouting (weekly), economic-threshold based sprays, and protection of beneficial insects.

Harvest and post-harvest

- Harvest when pods turn brown; dry and thresh.
- Dry produce to safe moisture before storage; use clean bags/bins and protect from rodents and insects.
- Keep records: seed lot, sowing date, inputs, irrigation, and yield—helps improve next season decisions.

Practical notes: Avoid late harvesting to reduce shattering; good option for drylands.

Appendix A: Fertilizer planning in simple steps

1) Get a soil test report (pH, OC, available N-P-K, S, Zn, B). 2) Choose a crop and target yield. 3) Use local recommendation as base and adjust for soil fertility. 4) Split nitrogen (and sometimes potassium) to reduce losses. 5) Combine organic manures with chemical fertilizers for better efficiency.

Example (conceptual): calculating urea requirement

If a recommendation says **60 kg N/ha** and you plan to supply it through urea (46% N), then urea needed ≈ 60/0.46 = 130 kg/ha. Split into 2-3 doses as advised for the crop stage.

Micronutrients: when to consider

- **Zinc (Zn):** common deficiency in rice and maize; apply zinc sulphate as per soil test / local advisory.
- **Sulphur (S):** important for oilseeds and pulses; SSP/gypsum can help where soils are S-deficient.
- **Boron (B):** affects seed setting in sunflower and mustard; apply only if deficient.

Appendix B: Practical water-saving tips

- **Level fields** to reduce water loss and improve uniformity (laser leveling where available).
- **Alternate wetting and drying (AWD)** in rice can reduce irrigation without yield loss in many systems (where suitable).
- **Drip and fertigation** in sugarcane and cotton can save water and improve nutrient-use efficiency.
- **Mulching** (crop residues, organic mulch) reduces evaporation and suppresses weeds.
- **Critical-stage irrigation** gives the highest return when water is limited.

Appendix C: Notes for interpreting crop groups in statistics

Agricultural tables often list crops by season (kharif/rabi/summer) and also show group totals. For example: **Cereals** may include rice, wheat, maize, barley and millets; **Nutri/Coarse cereals** refer broadly to millets and maize; **Total food grains** = cereals + pulses; **Total oilseeds** aggregates groundnut, soybean, rapeseed/mustard, sesame, sunflower, linseed, safflower, niger and others. Use these totals for macro planning, and the individual crop packages for field decisions.

Bale conversions (as commonly used in reports)

Cotton	1 bale = 170 kg (as per note provided)
Jute, sannhemp & mesta	1 bale = 180 kg (as per note provided)